THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · EASY LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. On simplifying $2x + 3 + 4x + 5$ we get: 1

(a) $8x + 8$ **(b)** $6x + 15$ **(c)** $14x$ **(d)** $6x + 8$

2. The area of a parallelogram with base 9 cm and height 4 cm is: 1

(a) 18 cm² **(b)** 36 cm² **(c)** 26 cm² **(d)** 13 cm²

3. The value of 4^3 is: 1

(a) 68 **(b)** 48 **(c)** 64 **(d)** 12

4. The value of 2^{-4} is: 1

(a) 16 **(b)** $\frac{1}{16}$ **(c)** -16 **(d)** $\frac{1}{8}$

5. The number 5 in binary is: 1

(a) 110 **(b)** 100 **(c)** 111 **(d)** 101

6. A rhombus has each side 6 cm. Its perimeter is: 1

(a) 24 cm **(b)** 12 cm **(c)** 36 cm **(d)** 18 cm

7. The sum of the digits of 2748 is: 1

(a) 20 **(b)** 23 **(c)** 21 **(d)** 22

8. $(a + b)^2$ equals: 1

(a) $2a + 2b$ **(b)** $a^2 + b^2$ **(c)** $a^2 + ab + b^2$ **(d)** $a^2 + 2ab + b^2$

9. The ratio of 50 cm to 2 m is: 1

(a) 4 : 1 **(b)** 2 : 4 **(c)** 1 : 4 **(d)** 1 : 5

10. The price ₹ 500 is increased by 10%. The new price (in ₹) is: 1

(a) 495 **(b)** 605 **(c)** 1100 **(d)** 550

11. The perimeter of a right triangle with legs 3 cm and 4 cm is (in cm): 1

(a) 7 **(b)** 10 **(c)** 12 **(d)** 13

12. On a map, 1 cm represents 5 km. A distance of 7 cm on the map is actually (in km): 1

(a) 12 **(b)** 35 **(c)** 40 **(d)** 70

13. The number of edges of a triangular prism is: 1

(a) 5 **(b)** 9 **(c)** 6 **(d)** 8

14. The range of 36, 30, 20, 53, 50, 45 is: 1

(a) 39 **(b)** 33 **(c)** 53 **(d)** 73

15. The two-digit number with tens digit x and ones digit 2 is written as: 1

(a) $2x$ **(b)** $20 + x$ **(c)** $10x + 2$ **(d)** $x + 2$

16. The area of a trapezium with parallel sides 8 cm and 6 cm and height 5 cm is: 1

(a) 120 cm² **(b)** 70 cm² **(c)** 40 cm² **(d)** 35 cm²

17. $\sqrt[3]{64}$ is: 1

(a) 3 **(b)** 5 **(c)** 6 **(d)** 4

18. 3200 in scientific notation is: 1

(a) 3.2×10^3 **(b)** 3.2×10^2 **(c)** 3.2×10^4 **(d)** 32×10^3

19. The place value of the digit 6 in 3652 is: 1

(a) 60 **(b)** 6000 **(c)** 6 **(d)** 600

20. A parallelogram has adjacent sides 8 cm and 5 cm. Its perimeter is: 1

(a) 13 cm **(b)** 26 cm **(c)** 40 cm **(d)** 21 cm

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Write an algebraic expression for: the sum of 7 and 5 times a number n. 2

22. Find the area of a rectangle of length 25 m and breadth 12 m. 2

23. Is 225 a perfect square? Give a reason. 2

24. Write 2^{-6} as a fraction. 2

25. Convert 111_2 to a decimal number. 2

26. One angle of a parallelogram is $48°$. Find the other three angles. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Write 4086 in expanded form. 3

28. The diagonals AC and BD of a rectangle ABCD meet at O. If AC = 16 cm, find BD, OA and OB. 3

29. Replace * by every possible digit so that 2*41 is divisible by 3. 3

30. Expand $(x + 2)(x + 3)$. 3

31. Check whether 3 : 4 and 15 : 20 are in proportion. 3

OR

The marked price of a bag is ₹ 1200. A discount of 10% is given. Find the discount and the selling price. 3

32. Find the perimeter of a right triangle whose legs are 5 cm and 12 cm. 3

OR

Divide ₹ 1200 among A, B and C in the ratio 2:3:5. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Find the smallest number that must be added to 4537 so that it becomes divisible by 9. 5

34. Find 104×25 using the distributive property and verify by direct multiplication. 5

35. 6 pens cost ₹ 90. Find the cost of 1 pen, the cost of 10 pens and the number of pens that can be bought for ₹ 210. 5

36. A trader buys a table for ₹ 2000. Find the selling price if he sells it at 5
(i) a gain of 10%
(ii) a loss of 5%. Also find the difference of the two selling prices.

OR

Using $(2m,\ m^2 - 1,\ m^2 + 1)$ with $m = 2$, write a Pythagorean triplet and verify it. 5

37. A camp has food for 20 people for 15 days. If 10 more people join the camp, for how many days will the same food last? 5

OR

In the Sierpinski triangle, stage 0 has 1 shaded triangle. Fill in the number of shaded triangles at stages 1, 2 and 3. By what factor does the count grow at each stage? 5

38. The mean of 5 numbers is 20. Four of them are 12, 18, 25, 30. Find the fifth number. 5

OR

In a number pyramid each block is the sum of the two blocks below it. The bottom row is 2, 5, 3. Find the middle row and the top block. 5

END OF QUESTION PAPER — **Total: 80 Marks**

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